Justin Mascarenhas

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Summary

Robotics engineer (M.S. Computer Engineering, RIT, 2026) who builds robots end to end: perception on NVIDIA Jetson, kinematics and planning, and the firmware that drives the joints. Delivered a voice-commanded, vision-guided pick-and-place arm with 100% pick reliability.

Skills

Robotics: ROS 2 / ROS, MoveIt, RViz, URDF, Forward & inverse kinematics, DH parameters, Visual servoing, Odometry, CARLA
Perception & Edge AI: YOLO, OpenCV, TensorRT, NVIDIA Jetson, PyTorch
Embedded & Control: Arduino, STM32, PWM servo & motor control, Quadrature encoders, Closed-loop feedback control, UART / I2C
Languages & Tools: C/C++, Python, Linux, systemd, Git, SolidWorks

Experience

General Secretary | Agnel Robotics Club (ARC), F.C.R.I.T. Jul 2023 – May 2024

- Trained a YOLOv3 ball detector from scratch and turned its detections into motor commands for closed-loop visual servoing of the competition robot.
- Led a 25-member robotics team through national-level competitions, owning mechanical design, PCB fabrication and AI integration end to end.

Research Intern (Capstone) | Bhabha Atomic Research Centre (BARC) Jul 2023 – Apr 2024

- Deployed ZoeDepth monocular depth estimation for crack-depth profiling with ±8% error on 3D surface reconstruction.

Projects

Voice-Controlled Robotic Arm (COFFIN) — M.S. Capstone, RIT Apr 2026

- Architected a voice-controlled, vision-guided pick-and-place robot as a three-layer distributed system: Flutter mobile client, Flask-SocketIO edge server, and an NVIDIA Jetson + Arduino controller on one real-time WebSocket fabric.
- Integrated faster-whisper speech-to-text (1.0–1.4 s), Gemini intent classification (<500 ms) and TensorRT-optimized YOLO for visual servoing, accelerating detection 5.2× over eager PyTorch (56.7 → 10.9 ms, 17.6 → 92 FPS).
- Implemented two-link inverse kinematics in C++ on the Arduino, driving five servos with contact-aware grasping from dual time-of-flight sensors on an I2C multiplexer: a camera range gate plus a gripper that closes at 25 mm.
- Drove the system to 100% pick reliability (10/10 trials) at ˜11.5 s voice-to-pick by hardening it against false-positive detections, servo brown-out and stale-frame capture.

ROS-Based Robotic Arm (AL5B & UR5e)

- Derived DH tables and full forward-kinematic chains by hand for a 6-DoF UR5e and a 5-DoF AL5B, validated against MoveIt and RViz URDF models in ROS 2 Humble and ROS Noetic.
- Planned and executed collision-free pick-and-place on the physical UR5e through the UR ROS 2 driver; AL5B repeatability under 2 mm over 10 trials.

Autonomous Driving in CARLA

- Trained YOLO11n (mAP50 ˜0.85) and fine-tuned UFLD (ResNet-18), integrated into an adaptive control stack with lane following and waypoint fallback.

Robocon Competition Robot — Holonomic Drive & Encoder Odometry 2022 – 2024

- Built closed-loop heading correction for a four-wheel holonomic robot on Arduino Due, deriving heading from dual-axis quadrature-encoder odometry (2400 CPR) and feeding the angular error back into rotation PWM.

Education

M.S., Computer Engineering | Rochester Institute of Technology Aug 2024 – May 2026
Courses: Principles of Robotics, Robot Perception, Machine Intelligence, Deep Learning

B.E., Electronics & Telecommunication | University of Mumbai (F.C.R.I.T.) 2020 – 2024
Courses: Image Processing, Digital Signal Processing